DISCUSSION DOCUMENT - Assignment 3

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This assignment has two parts. Part 1 builds Hidden Markov Model (HMM) classifiers — GaussianHMM and MultinomialHMM — for the UCI Ionosphere and Breast Cancer datasets, with one class-wise HMM trained per label and predictions made by likelihood comparison. Parts 2 & 3 compare five deep learning architectures — CNN, VGG16-style, AlexNet-style, GoogLeNet-style, and an RNN on image rows — on MNIST and CIFAR-10.

Models are evaluated with and without hyperparameter tuning, across multiple train-test splits, using Accuracy, Precision, Recall, F1, confusion matrices, training/loss curves, and ROC-AUC.

Assignment Link: Google Colab notebook

1. Datasets

Ionosphere (Part 1): 351 samples, 34 continuous features, binary (126 vs 225).

Breast Cancer (Part 1): 569 samples, 30 numeric features, binary (212 malignant vs 357 benign).

MNIST (Parts 2 & 3): 28x28 grayscale digits, 10 classes, 12,000 samples used with custom splits.

CIFAR-10 (Parts 2 & 3): 32x32 RGB natural images, 10 classes, 12,000 samples used with custom splits.

Preprocessing: for Part 1, each feature vector is reshaped into a short sequence for the HMM; MultinomialHMM additionally discretizes continuous features into bins. For Parts 2 & 3, pixels are normalized and labels one-hot encoded.

2. Evaluation Metrics

- Accuracy — percentage of correctly classified samples.
- Precision — fraction of predicted positives that are correct.
- Recall — fraction of actual positives correctly identified.
- F1-score — harmonic mean of Precision and Recall.
- Confusion Matrix — per-class correct/incorrect prediction counts.
- ROC-AUC — ability of the classifier to separate classes.

3. Part 1 — Hidden Markov Model (HMM) Classification

Each sample's feature vector is treated as a short observation sequence; one HMM is trained per class via Baum-Welch (EM). A test sample is assigned to the class-HMM with the highest log-likelihood.

3.1 GaussianHMM

Where it succeeds	Where it fails
• Handles continuous features natively, no information loss.	• Sensitive to n_components / covariance_type choice.
• Achieved the higher accuracy of the two variants on both datasets.	• Can show convergence warnings on small/noisy data.

3.2 MultinomialHMM

Where it succeeds	Where it fails
• Simple emission model, fewer parameters per state.	• Discretization into bins loses information — far lower accuracy.
• Faster to train in some configurations.	• Predictions sometimes collapsed to one class (0 precision/recall).

3.3 Hyperparameter Tuning

Grid search over n_components (1–4) and, for GaussianHMM, covariance_type (diag/full), selected via a validation split.

Where it succeeds	Where it fails
• Consistently improved or matched GaussianHMM's default accuracy.	• Little benefit for MultinomialHMM — its core limit is discretization, not n_components.
• Best overall Part-1 accuracy (91.2%) came from a tuned GaussianHMM.	• Adds noticeable training time across 4 splits × 2 datasets × 2 classifiers.

4. Part 1 Results — HMM

4.1 Full Results — All Splits, Untuned vs Tuned

HMM Results — All Splits, Untuned vs Tuned

Dataset	Classifier	Tuned	Split	Params	Acc	Prec	Rec	F1
BreastCancer	GaussianHMM	No	0.6	default	0.8640	0.9444	0.8322	0.8848
BreastCancer	GaussianHMM	No	0.7	default	0.8655	0.9468	0.8318	0.8856
BreastCancer	GaussianHMM	No	0.8	default	0.8596	0.9667	0.8056	0.8788
BreastCancer	GaussianHMM	No	0.9	default	0.8772	0.9677	0.8333	0.8955
BreastCancer	GaussianHMM	Yes	0.6	nc=1,diag	0.8860	0.8926	0.9301	0.9110
BreastCancer	GaussianHMM	Yes	0.7	nc=1,diag	0.8947	0.9083	0.9252	0.9167
BreastCancer	GaussianHMM	Yes	0.8	nc=3,diag	0.8509	0.9661	0.7917	0.8702
BreastCancer	GaussianHMM	Yes	0.9	nc=4,diag	0.9123	1.0000	0.8611	0.9254
BreastCancer	MultinomialHMM	No	0.6	default	0.3728	0.0000	0.0000	0.0000
BreastCancer	MultinomialHMM	No	0.7	default	0.6257	0.6257	1.0000	0.7698
BreastCancer	MultinomialHMM	No	0.8	default	0.3684	0.0000	0.0000	0.0000
BreastCancer	MultinomialHMM	No	0.9	default	0.6316	0.6316	1.0000	0.7742
BreastCancer	MultinomialHMM	Yes	0.6	nc=2	0.3728	0.0000	0.0000	0.0000
BreastCancer	MultinomialHMM	Yes	0.7	nc=4	0.6257	0.6257	1.0000	0.7698
BreastCancer	MultinomialHMM	Yes	0.8	nc=3	0.6316	0.6316	1.0000	0.7742
BreastCancer	MultinomialHMM	Yes	0.9	nc=3	0.6316	0.6316	1.0000	0.7742
Ionosphere	GaussianHMM	No	0.6	default	0.7730	0.8372	0.8000	0.8182
Ionosphere	GaussianHMM	No	0.7	default	0.7642	0.8308	0.7941	0.8120
Ionosphere	GaussianHMM	No	0.8	default	0.8310	0.9048	0.8261	0.8636
Ionosphere	GaussianHMM	No	0.9	default	0.7500	0.8500	0.7391	0.7907
Ionosphere	GaussianHMM	Yes	0.6	nc=2,diag	0.7730	0.8372	0.8000	0.8182
Ionosphere	GaussianHMM	Yes	0.7	nc=3,diag	0.7925	0.7738	0.9559	0.8553
Ionosphere	GaussianHMM	Yes	0.8	nc=4,diag	0.8873	0.9318	0.8913	0.9111
Ionosphere	GaussianHMM	Yes	0.9	nc=4,diag	0.8333	0.9048	0.8261	0.8636
Ionosphere	MultinomialHMM	No	0.6	default	0.6383	0.6383	1.0000	0.7792
Ionosphere	MultinomialHMM	No	0.7	default	0.6415	0.6415	1.0000	0.7816
Ionosphere	MultinomialHMM	No	0.8	default	0.3521	0.0000	0.0000	0.0000
Ionosphere	MultinomialHMM	No	0.9	default	0.3611	0.0000	0.0000	0.0000
Ionosphere	MultinomialHMM	Yes	0.6	nc=1	0.3617	0.0000	0.0000	0.0000
Ionosphere	MultinomialHMM	Yes	0.7	nc=2	0.6415	0.6415	1.0000	0.7816
Ionosphere	MultinomialHMM	Yes	0.8	nc=3	0.3521	0.0000	0.0000	0.0000
Ionosphere	MultinomialHMM	Yes	0.9	nc=2	0.3611	0.0000	0.0000	0.0000

Screenshot of the notebook's Cell 9 output: GaussianHMM vs MultinomialHMM, default vs tuned, across all four splits.

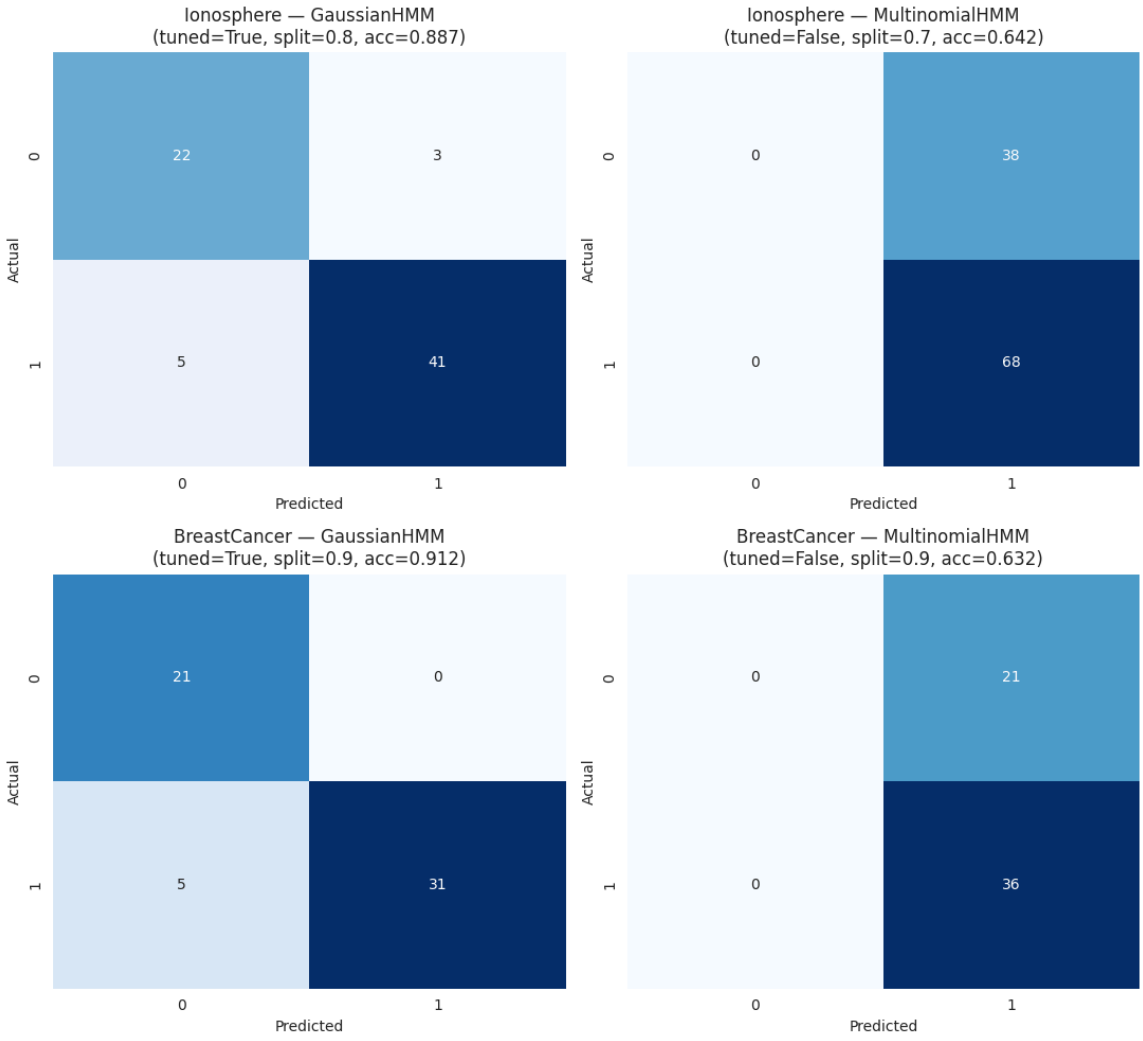
4.2 Best Configuration per Dataset & Classifier

Best Configuration per Dataset & Classifier

Dataset	Classifier	Tuned	Split	Params	Acc	Prec	Rec	F1
BreastCancer	GaussianHMM	Yes	0.9	nc=4,diag	0.9123	1.0000	0.8611	0.9254
BreastCancer	MultinomialHMM	No	0.9	default	0.6316	0.6316	1.0000	0.7742
Ionosphere	GaussianHMM	Yes	0.8	nc=4,diag	0.8873	0.9318	0.8913	0.9111
Ionosphere	MultinomialHMM	No	0.7	default	0.6415	0.6415	1.0000	0.7816

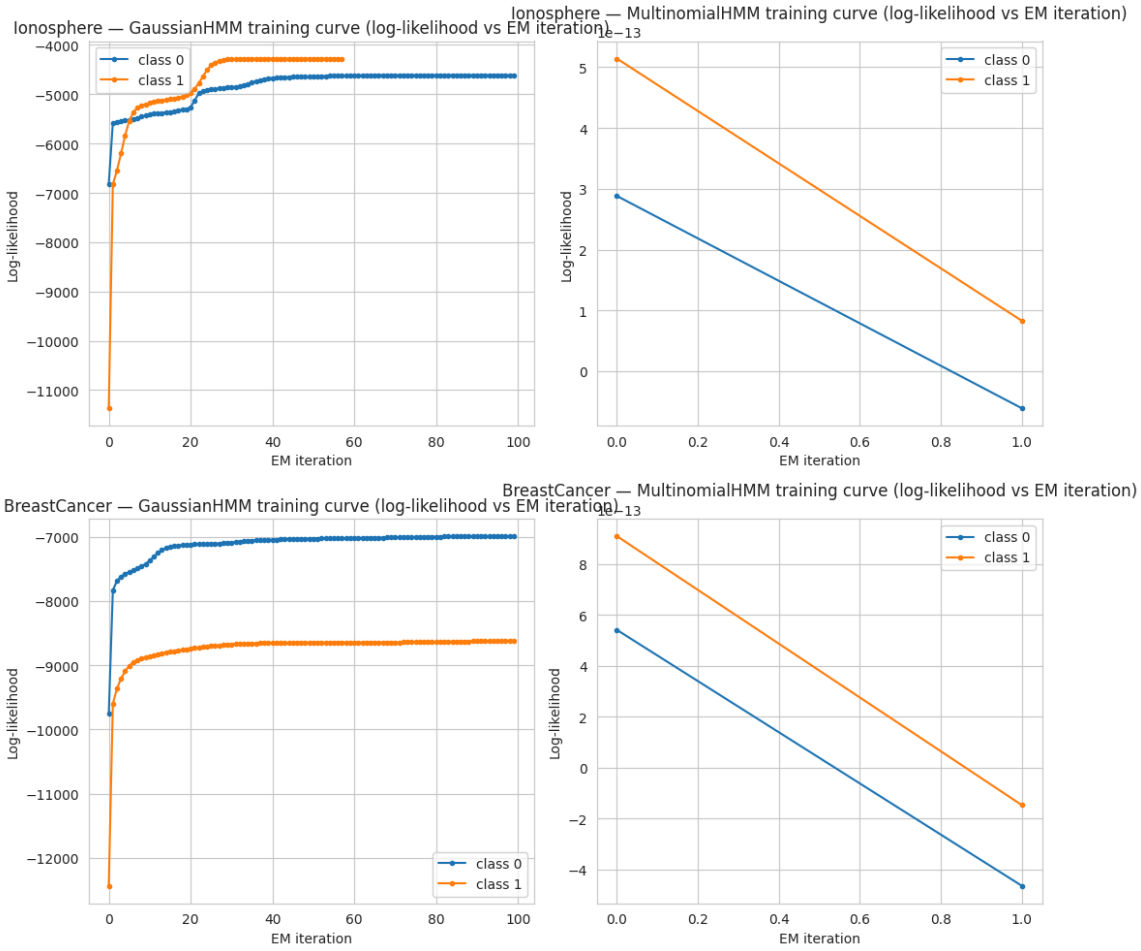
Screenshot of Cell 10 output — highest-accuracy configuration per (dataset, classifier).

4.3 Confusion Matrices (Best Case)

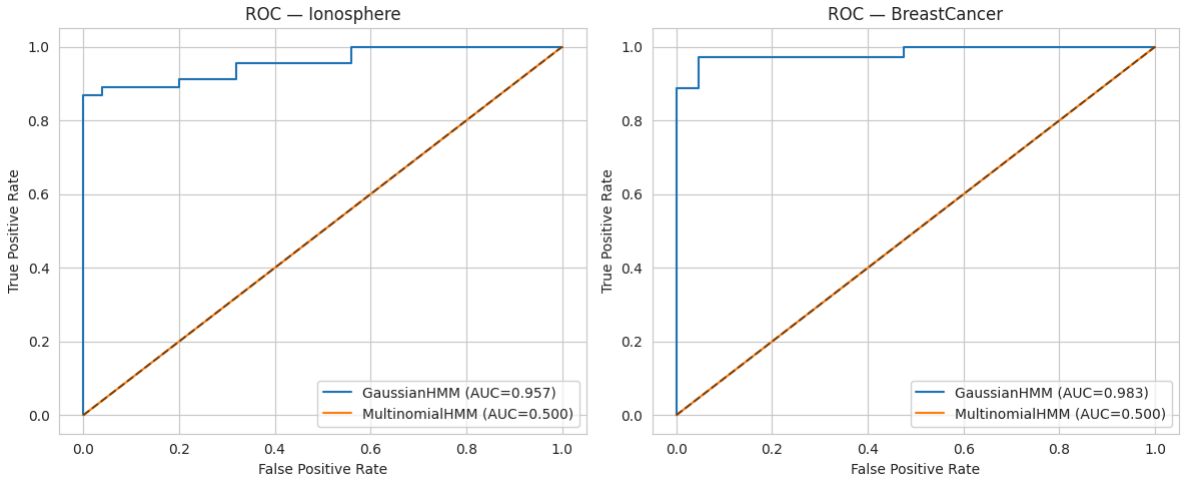


Confusion matrices for the best configuration of each classifier on Ionosphere and Breast Cancer.

4.4 Training (Log-Likelihood) Curves & ROC-AUC



EM log-likelihood convergence — GaussianHMM converges smoothly; MultinomialHMM (2-state) converges almost immediately.



ROC curves: GaussianHMM AUC = 0.957 (Ionosphere), 0.983 (Breast Cancer); MultinomialHMM collapsed to AUC = 0.500 on these splits.

4.5 Final Accuracy Comparison by Split

Final Accuracy Comparison by Split — HMM

Dataset	Classifier	Tuned	0.6	0.7	0.8	0.9
BreastCancer	GaussianHMM	No	0.8640	0.8655	0.8596	0.8772
BreastCancer	GaussianHMM	Yes	0.8860	0.8947	0.8509	0.9123
BreastCancer	MultinomialHMM	No	0.3728	0.6257	0.3684	0.6316
BreastCancer	MultinomialHMM	Yes	0.3728	0.6257	0.6316	0.6316
Ionosphere	GaussianHMM	No	0.7730	0.7642	0.8310	0.7500
Ionosphere	GaussianHMM	Yes	0.7730	0.7925	0.8873	0.8333
Ionosphere	MultinomialHMM	No	0.6383	0.6415	0.3521	0.3611
Ionosphere	MultinomialHMM	Yes	0.3617	0.6415	0.3521	0.3611

5. Parts 2 & 3 — Deep Learning Model Comparison

Five architectures — CNN, VGG16-style, AlexNet-style, GoogLeNet-style, and an RNN on image rows — were trained for 12 epochs (batch size 128) on MNIST and CIFAR-10, at train fractions 0.6/0.7/0.8 (30 experiments total, checkpointed to Drive).

5.1 CNN (baseline)

Where it succeeds	Where it fails
• Simple, fast, strong MNIST baseline (~97.5%).	• Limited capacity for CIFAR-10 (~53–58%).
• Easy to interpret and debug.	• No explicit depth/regularization beyond dropout.

5.2 VGG16-style

Where it succeeds	Where it fails
• Deep small-filter stack; best/near-best on both datasets.	• More parameters and compute than plain CNN.
• MNIST ~98.0%, CIFAR-10 ~59%.	• Slower per-epoch training.

5.3 AlexNet-style

Where it succeeds	Where it fails
• Strongest CIFAR-10 performer (~62%).	• Large early filters lose fine detail vs deeper small-filter nets.
• Also ties for best MNIST accuracy (~97.9%).	• Prone to overfitting with limited data/epochs.

5.4 GoogLeNet-style (Inception)

Where it succeeds	Where it fails
• Multi-branch modules capture multi-scale features.	• Weakest CNN-family model on CIFAR-10 (~46–51%).
• Parameter-efficient relative to its representational power.	• Slower convergence within 12 epochs.

5.5 RNN (on image rows)

Where it succeeds	Where it fails
• Shows how a sequence model handles images as row-sequences.	• Weakest model overall (CIFAR-10 ~41–46%).
• Still reaches ~93–95% on MNIST.	• Row-wise sequences discard 2D spatial locality.

5.6 Training Setup

EPOCHS=12, BATCH_SIZE=128, splits {0.6, 0.7, 0.8}. Training/validation accuracy & loss logged every epoch; checkpointing made the sweep resumable after Colab disconnects. No LR scheduling or data augmentation was used, which likely limits CIFAR-10 accuracy.

6. Parts 2 & 3 Results — Deep Learning

6.1 Full Results — MNIST

Deep Learning Results — MNIST (all models & splits)

Dataset	Model	Split	Accuracy	Precision	Recall	F1
MNIST	CNN	0.6	0.9706	0.9705	0.9706	0.9703
MNIST	CNN	0.7	0.9731	0.9731	0.9729	0.9729
MNIST	CNN	0.8	0.9746	0.9751	0.9742	0.9745
MNIST	VGG16	0.6	0.9704	0.9702	0.9706	0.9701
MNIST	VGG16	0.7	0.9797	0.9796	0.9797	0.9796
MNIST	VGG16	0.8	0.9792	0.9792	0.9791	0.9791
MNIST	AlexNet	0.6	0.9783	0.9781	0.9784	0.9782
MNIST	AlexNet	0.7	0.9789	0.9788	0.9789	0.9788
MNIST	AlexNet	0.8	0.9792	0.9795	0.9791	0.9791
MNIST	GoogLeNet	0.6	0.9602	0.9604	0.9597	0.9599
MNIST	GoogLeNet	0.7	0.9683	0.9688	0.9681	0.9683
MNIST	GoogLeNet	0.8	0.9667	0.9671	0.9666	0.9666
MNIST	RNN	0.6	0.9333	0.9343	0.9330	0.9331
MNIST	RNN	0.7	0.9425	0.9429	0.9418	0.9420
MNIST	RNN	0.8	0.9454	0.9460	0.9449	0.9450

6.2 Full Results — CIFAR-10

Deep Learning Results — CIFAR-10 (all models & splits)

Dataset	Model	Split	Accuracy	Precision	Recall	F1
CIFAR10	CNN	0.6	0.5350	0.5403	0.5348	0.5351
CIFAR10	CNN	0.7	0.5794	0.5878	0.5796	0.5759
CIFAR10	CNN	0.8	0.5571	0.5630	0.5580	0.5498
CIFAR10	VGG16	0.6	0.5408	0.5623	0.5406	0.5436
CIFAR10	VGG16	0.7	0.5919	0.5976	0.5917	0.5874
CIFAR10	VGG16	0.8	0.5796	0.5905	0.5795	0.5772
CIFAR10	AlexNet	0.6	0.5869	0.5833	0.5864	0.5815
CIFAR10	AlexNet	0.7	0.5983	0.6079	0.5971	0.5973
CIFAR10	AlexNet	0.8	0.6204	0.6327	0.6201	0.6208
CIFAR10	GoogLeNet	0.6	0.4585	0.4755	0.4580	0.4480
CIFAR10	GoogLeNet	0.7	0.5075	0.4987	0.5065	0.4917
CIFAR10	GoogLeNet	0.8	0.5096	0.5335	0.5109	0.5030
CIFAR10	RNN	0.6	0.4081	0.4216	0.4077	0.4059
CIFAR10	RNN	0.7	0.4386	0.4357	0.4388	0.4233
CIFAR10	RNN	0.8	0.4550	0.4569	0.4543	0.4471

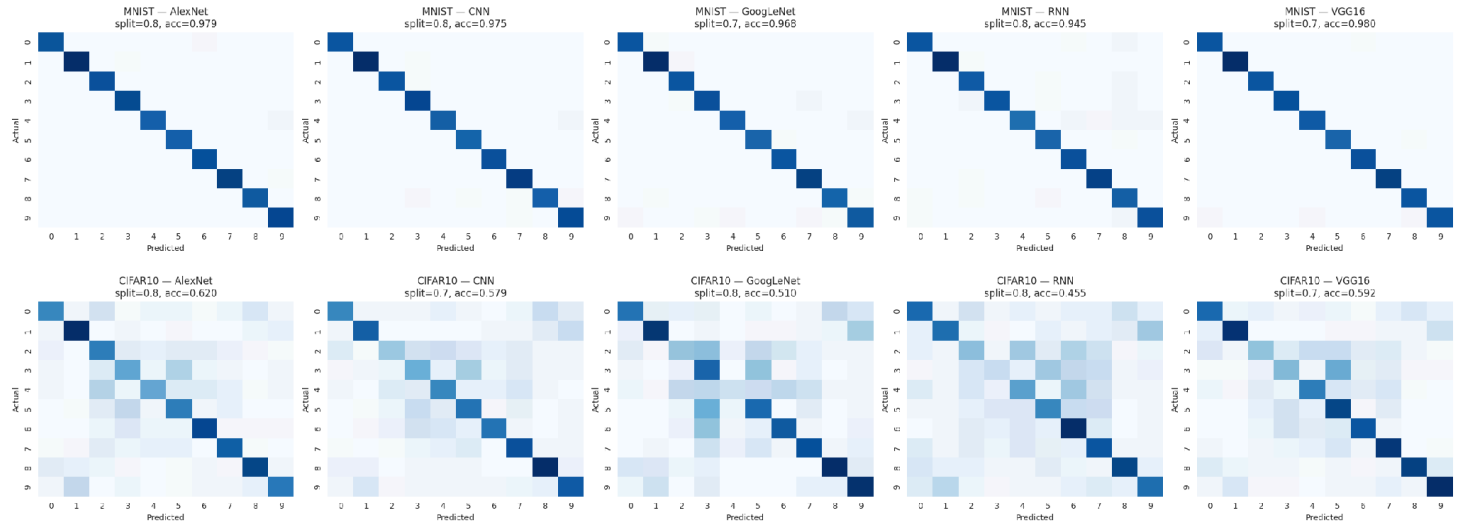
Screenshots of the notebook's Cell 6 output (results table) for MNIST and CIFAR-10 respectively.

6.3 Best Case per Dataset & Model

Best Case per Dataset & Model						
Dataset	**Model**	**Split**	**Accuracy**	**Precision**	**Recall**	**F1**
MNIST	AlexNet	0.8	0.9792	0.9795	0.9791	0.9791
MNIST	VGG16	0.7	0.9797	0.9796	0.9797	0.9796
MNIST	CNN	0.8	0.9746	0.9751	0.9742	0.9745
MNIST	GoogLeNet	0.7	0.9683	0.9688	0.9681	0.9683
MNIST	RNN	0.8	0.9454	0.9460	0.9449	0.9450
CIFAR10	AlexNet	0.8	0.6204	0.6327	0.6201	0.6208
CIFAR10	VGG16	0.7	0.5919	0.5976	0.5917	0.5874
CIFAR10	CNN	0.7	0.5794	0.5878	0.5796	0.5759
CIFAR10	GoogLeNet	0.8	0.5096	0.5335	0.5109	0.5030
CIFAR10	RNN	0.8	0.4550	0.4569	0.4543	0.4471

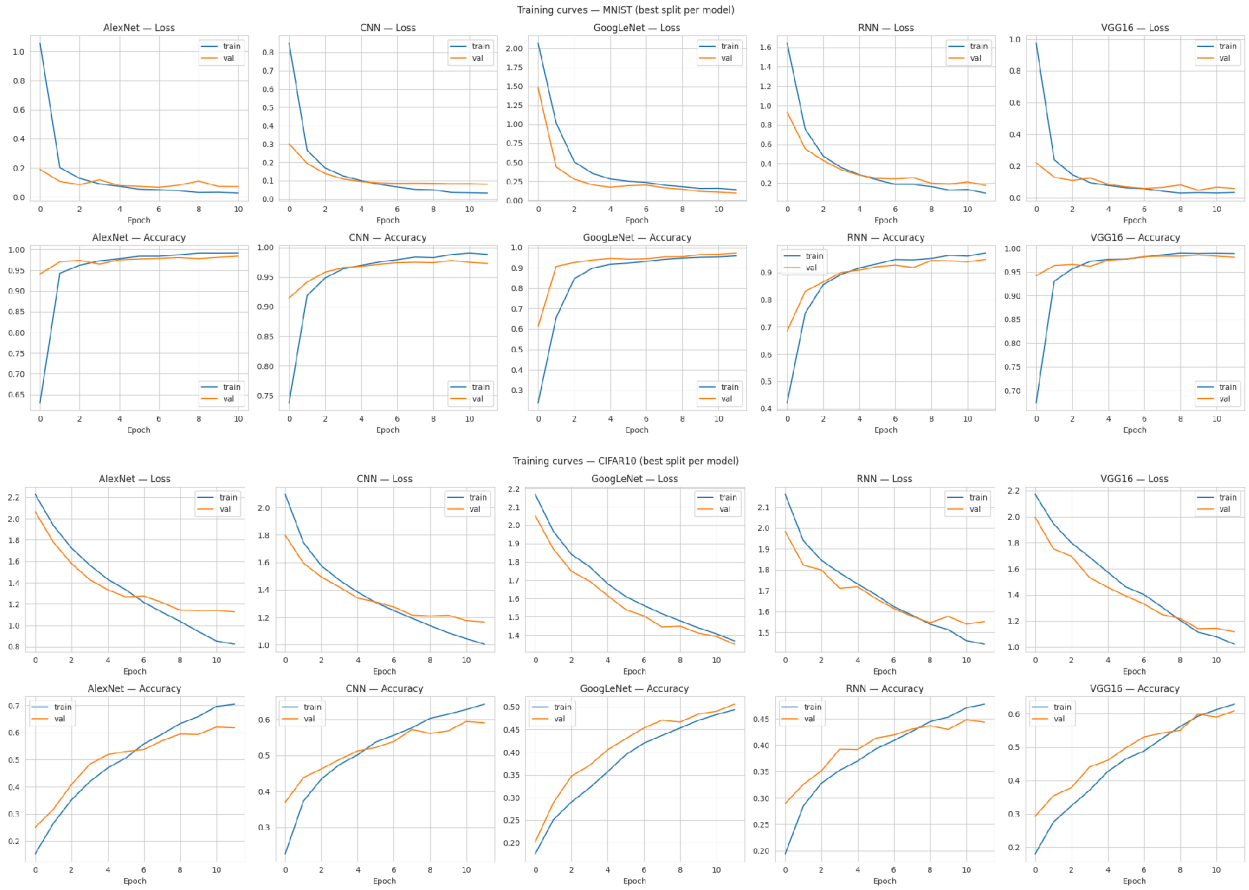
Ranked by accuracy — MNIST: VGG16 ≈ AlexNet > CNN > GoogLeNet > RNN. CIFAR-10: AlexNet > VGG16 > CNN > GoogLeNet > RNN.

6.4 Confusion Matrices (Best Case per Model)



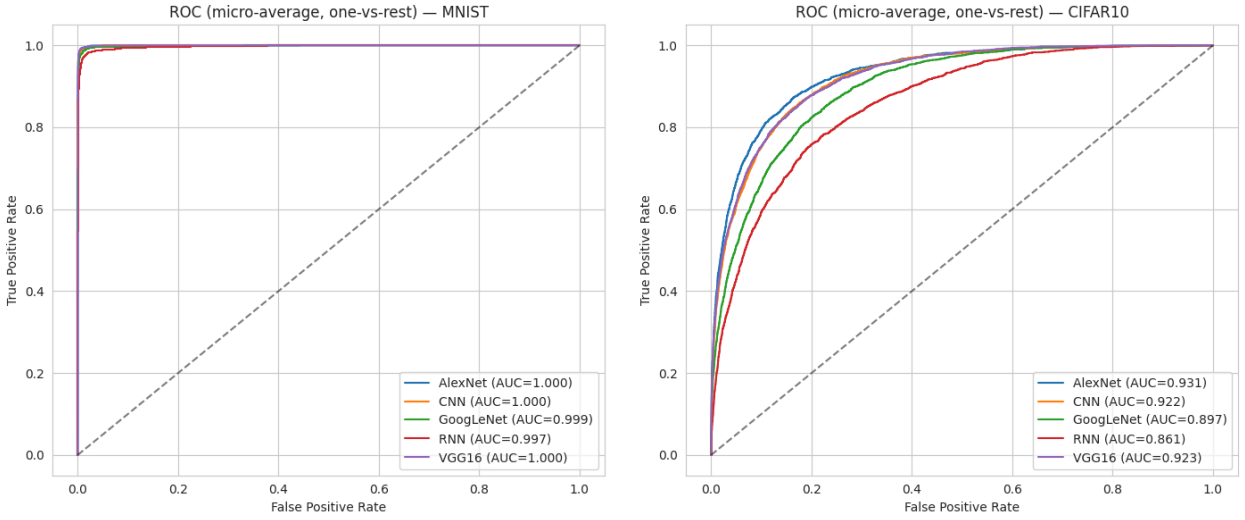
Top: MNIST. Bottom: CIFAR-10. Diagonal-dominant on MNIST; far more confusion on CIFAR-10, especially GoogLeNet and RNN.

6.5 Training & Validation Loss / Accuracy Curves



Top: MNIST — fast convergence, little train/val gap. Bottom: CIFAR-10 — slower convergence with a widening train/val gap for AlexNet and VGG16 (early overfitting).

6.6 ROC-AUC (micro-average, one-vs-rest)



Left: MNIST — all models near-perfect (AUC 0.997–1.000). Right: CIFAR-10 — AUC spreads 0.861–0.931, AlexNet highest, RNN lowest.

6.7 Final Accuracy Comparison by Split

Final Accuracy Comparison by Split — Deep Learning				
Dataset	**Model**	**0.6**	**0.7**	**0.8**
MNIST	CNN	0.9706	0.9731	0.9746
MNIST	VGG16	0.9704	0.9797	0.9792
MNIST	AlexNet	0.9783	0.9789	0.9792
MNIST	GoogLeNet	0.9602	0.9683	0.9667
MNIST	RNN	0.9333	0.9425	0.9454
CIFAR10	CNN	0.5350	0.5794	0.5571
CIFAR10	VGG16	0.5408	0.5919	0.5796
CIFAR10	AlexNet	0.5869	0.5983	0.6204
CIFAR10	GoogLeNet	0.4585	0.5075	0.5096
CIFAR10	RNN	0.4081	0.4386	0.4550

7. Discussion

Part 1 (HMM): GaussianHMM clearly outperforms MultinomialHMM on both datasets since their features are continuous and discretization discards information (occasionally collapsing MultinomialHMM to one-class predictions). Tuning n_components/covariance_type generally helped GaussianHMM, with the best results at larger train fractions (0.8–0.9) and 3–4 hidden components. Accuracy improved with more training data, though not perfectly monotonically — reflecting split-to-split variance on datasets this size.

Parts 2 & 3 (Deep Learning): On MNIST all five models score strongly (93–98%), with VGG16/AlexNet ahead of CNN/GoogLeNet and the RNN trailing. On CIFAR-10 the gap widens sharply: AlexNet and VGG16 lead (~59–62%), CNN/GoogLeNet trail (~46–58%), and the RNN is weakest (~41–46%) since treating image rows as a 1D sequence discards the 2D spatial locality convolutions exploit. Increasing train fraction from 0.6→0.8 improved most models on both datasets; CIFAR-10 curves converge more slowly and show early train/val divergence within 12 epochs, suggesting more epochs, augmentation, or LR scheduling would help further.

8. Conclusion

Class-wise GaussianHMM and MultinomialHMM classifiers were built, tuned and evaluated on Ionosphere and Breast Cancer; tuned GaussianHMM was the clear winner (up to 91.2% / 88.7% accuracy). Five deep learning architectures were compared on MNIST and CIFAR-10 across three splits; AlexNet and VGG16 were strongest overall (~98% MNIST, ~59–62% CIFAR-10), while the RNN-on-rows approach was consistently weakest — confirming convolutional architectures suit image data better than a naively-applied sequence model. Across both parts, model choice, hyperparameter tuning, and more training data each meaningfully improved performance.